Long Short-Term Memory Networks for Predicting Pore Pressure in Liquefaction

Krishna Kumar & Ellen Rathje

University of Texas at Austin

Overview

What is Liquefaction?

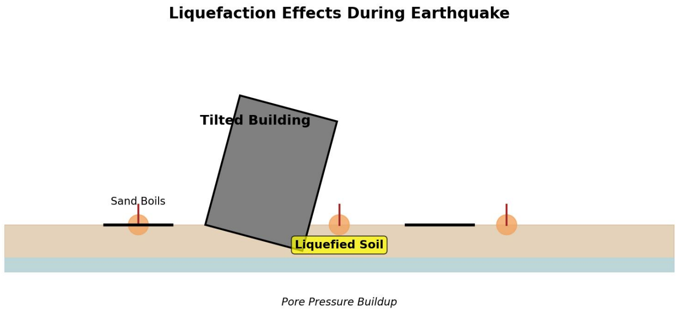


Fig. Liquefaction damage during earthquake

Definition: Saturated soil loses strength during cyclic loading (earthquakes)

Consequences:

- Building settlements and tilting
- Bridge foundation failures

Effective Stress Principle:

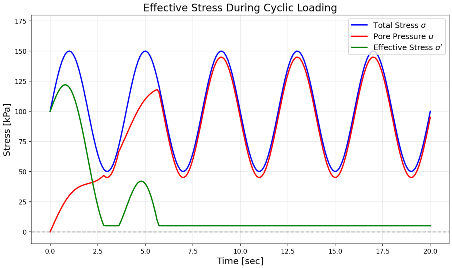
$$\sigma' = \sigma - u \quad (1)$$

where σ' = effective stress, σ = total stress, u = pore water pressure

Pore Pressure Ratio:

$$r_u = \frac{u_{excess}}{\sigma'_0} \quad (2)$$

Liquefaction occurs when: $r_u \geq 1.0$

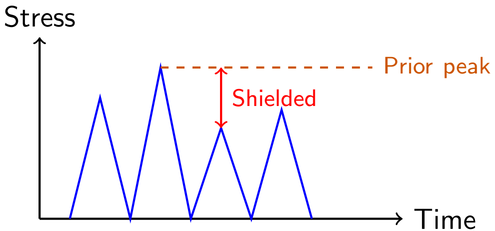


The Challenge: Stress History Dependency

Key Observation: Pore pressure generation depends on **entire stress history**, not just current stress!

The Shielding Effect:

- When current shear stress amplitude < prior peak amplitude
- Pore pressure remains nearly constant ("shielded")
- Sophisticated constitutive models (PM4Sand, PDMY02) **fail** to capture this



Limitations of Constitutive Models

Common Models: PM4Sand, PDMY02, Dafalias-Manzari

Problems:

1. Cannot capture shielding effect
2. Predict continuous pore pressure increase
3. Miss stress history dependencies
4. Poor predictions of liquefaction timing
5. Require extensive calibration (15-30+ parameters)

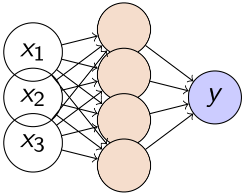
Why Simple Neural Networks Fail Too:

- Take current state as input: $y_t = f(x_t)$
- **No memory** of past events
- Cannot learn temporal dependencies

What We Need: $y_t = f(x_t, x_{t-1}, x_{t-2}, \ldots, x_0)$

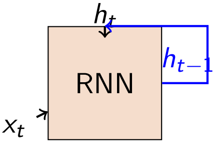
From Feedforward to Recurrent Networks

Feedforward Neural Network:



No memory, fixed input size

Recurrent Neural Network:



Loops allow information persistence

RNN Equation:

$$h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \quad (3)$$

The Vanishing Gradient Problem

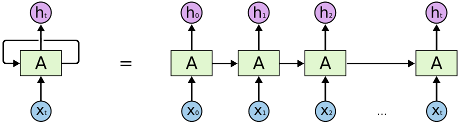
Problem with RNNs:

- Gradients multiplied many times during backpropagation
- Gradients vanish ($\to 0$) or explode ($\to \infty$)
- Cannot learn long-term dependencies

$$\frac{\partial L}{\partial W} = \prod_{t=1}^{T} \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_t}{\partial W} \tag{4}$$

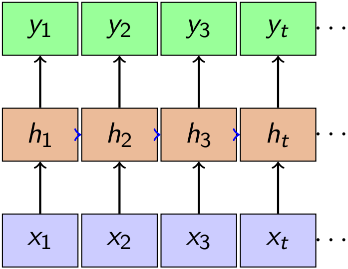
If $\left|\frac{\partial h_t}{\partial h_{t-1}}\right| < 1$: gradient $\to 0$ as $T \to \infty$

Impact: Can't remember events 10+ steps ago
For liquefaction: Need to remember 100s-1000s of loading cycles!



Sequence-to-Sequence Learning

General Framework: Map input sequence $\rightarrow$ output sequence



For Liquefaction:

- **Input sequence:** Stress history $[\tau_1, \tau_2, \ldots, \tau_t]$
- **Output:** Pore pressure ratio $r_u(t)$
- Use sliding window (window $=$ 800 timesteps $\approx$ 2 cycles)

Long Short-Term Memory (LSTM)

Key Innovation: Replace simple RNN cell with complex **memory cell**

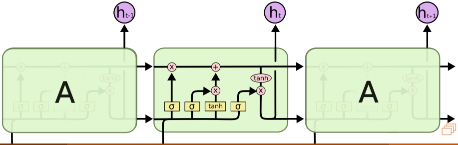
Developed by: Hochreiter & Schmidhuber (1997)

LSTM Can:

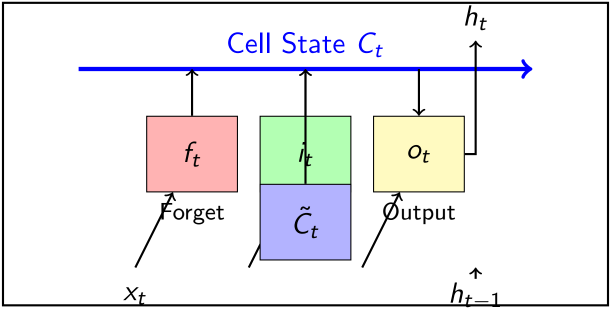
1. **Remember** information for long periods
2. **Forget** irrelevant information
3. **Selectively update** memory

Applications:

- Machine translation
- Speech recognition
- Time series forecasting
- **Now: Geotechnical engineering!**



LSTM Architecture: The Three Gates



Three gates control information flow:

1. Forget gate f_t: What to remove from memory
2. Input gate i_t: What new info to store
3. Output gate o_t: What to output now

LSTM Mathematics: Gate Equations

Input: x_t (current), h_{t-1} (previous hidden), C_{t-1} (previous cell state)

1. Forget Gate (what to discard):

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{5}$$

2. Input Gate (what to add):

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{6}$$

3. Candidate Cell State:

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{7}$$

Where: σ = sigmoid (outputs 0 to 1), tanh = hyperbolic tangent (outputs -1 to 1)

LSTM Mathematics: State Updates

4. Update Cell State:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{8}$$

- $f_t \odot C_{t-1}$: Forget old information
- $i_t \odot \tilde{C}_t$: Add new information
- $\odot$: Element-wise multiplication

5. Output Gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{9}$$

6. Hidden State:

$$h_t = o_t \odot \tanh(C_t) \tag{10}$$

Key Point: Cell state C_t flows with minimal modification $\rightarrow$ solves vanishing gradient!

Regular RNN:

- Gradient flows through tanh repeatedly
- After T steps: gradient ≈ 0

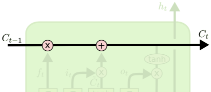
LSTM:

- Cell state provides **highway** for gradients
- Additive updates preserve gradient

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{11}$$

Analogy: LSTM is like a smart notebook

- Forget gate: Eraser (remove old notes)
- Input gate: Pen filter (decide what to write)
- Output gate: Cover (control what readers see)



Problem Formulation

Goal: Predict pore pressure ratio $r_u(t)$ from stress history

Input Features:

1. Normalized time: $T_{\text{norm}} = t/t_{\text{final}}$
2. Normalized shear stress: $\tau_{\text{norm}} = \tau/\sigma'_0$
3. Relative density: D_r (constant)

Output: Pore pressure ratio $r_u \in [0, 1]$

Time-Window Approach:

$$\mathbf{X}_t = \begin{bmatrix} T_t & D_r & \tau_t \\ T_{t+1} & D_r & \tau_{t+1} \\ \vdots & \vdots & \vdots \\ T_{t+799} & D_r & \tau_{t+799} \end{bmatrix}_{800\times 3} \quad \rightarrow \quad \mathbf{Y}_t = r_{u,t+800} \tag{12}$$

Window length: 800 timesteps $\approx$ 2 loading cycles

Network Architecture

Input
(800 × 3)

Time, D_r, Stress

LSTM Layer 1
128 units

Sequence processing

LSTM Layer 2
128 units

Stress history memory

Dense
64 units, tanh

Dense
16 units, tanh

Post-processing

Output
1 unit, tanh

r_u prediction

Total Parameters: 200,505

Why LSTM is Perfect for Liquefaction

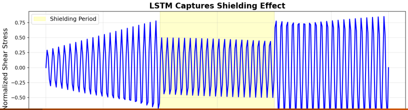
The Shielding Effect - Learned Automatically:

1. Forget gate: Discards low-stress periods
2. Input gate: Remembers peak stresses
3. Output gate: Controls r_u output
4. Cell state: Long-term stress memory

Learned Behavior:

- When $\tau_{\text{current}} > \tau_{\text{peak}}$:
  - Input gate opens $\rightarrow$ store new peak
  - r_u increases
- When $\tau_{\text{current}} < \tau_{\text{peak}}$:
  - Forget gate filters
  - r_u plateaus (shielding!)

No Hard-Coding! LSTM discovers "if current $<$ peak" logic from data



Experimental Data: Nevada Sand

Data Source: Kwan et al. (2017) — Cyclic Simple Shear Tests

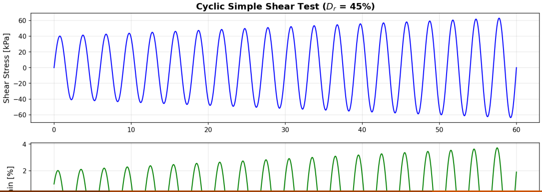
Test Conditions:

- Saturated Nevada sand
- Loose: $D_r = 33\text{-}55\%$ (Exp 9)
- Dense: $D_r = 72\text{-}94\%$ (Exp 8)

Loading Types: Harmonic, Modulated-up, Modulated-down

Training Set:

- 10 trials from Experiment 9
- ∼125,000 training samples
- 20% validation split



Training Strategy

Data Preprocessing:

- Augment with 900 static timesteps (initial state)
- Normalize time, stress, and D_r to [0, 1]

Memory Optimization:

- Sampling stride = 4 (75% memory reduction)
- File-by-file processing
- Explicit garbage collection
- float32 instead of float64

Hyperparameters:

- Optimizer: Adam, Learning rate: 0.0002
- Batch size: 32, Epochs: 50
- Loss: Mean Squared Error (MSE)
- Early stopping: patience = 1000

Hardware: Apple Silicon GPU (MPS), PyTorch 2.7, Training time: ~1 hour

Training Performance

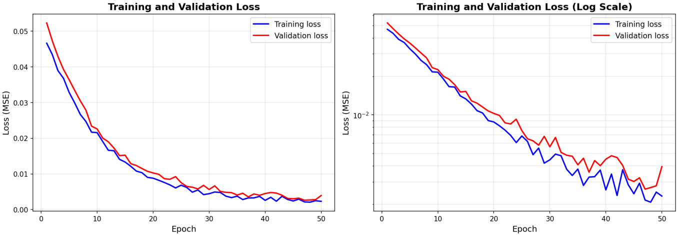


Fig. Training and validation loss over epochs

Observations:

- Rapid initial learning
- Validation tracks training (no overfitting)
- Smooth convergence

Test Set Predictions

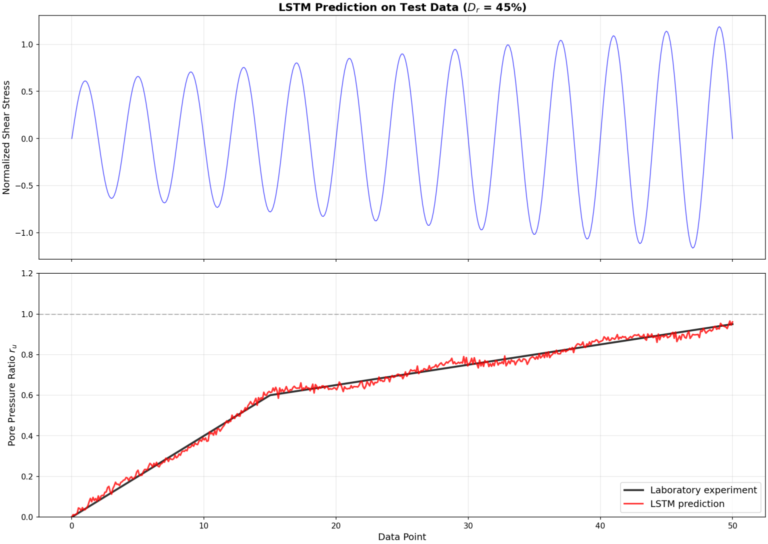


Fig. LSTM predictions vs experimental measurements on test data

Model Performance

Test Set Metrics:

Metric	Value
MSE	TBD
MAE	TBD
R^2	TBD

What LSTM Captures:

- ✓Shielding effect
- ✓Pore pressure buildup rate
- ✓Time to liquefaction
- ✓Density effects

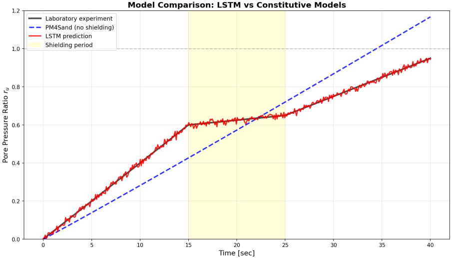


Fig. LSTM vs PM4Sand/PDMY02

vs Constitutive Models:

- PM4Sand: No shielding
- PDMY02: Overestimates r_u
- LSTM: Captures fluctuations

Key Findings

Success Factors:

1. Memory mechanism maintains stress history
2. Automatic feature learning (gates discover shielding logic)
3. Sequence modeling captures temporal context
4. Data-driven (no complex parameter calibration)

Advantages:

- ✓Captures stress history
- ✓No physics assumptions
- ✓Fast predictions
- ✓Generalizes well

Limitations:

- Requires training data
- Less transparent than physics
- Extrapolation uncertain
- Needs GPU for training

Future Directions

Model Improvements:

- Add attention mechanisms
- Incorporate physics constraints
- Multi-task learning (predict strain + r_u)
- Uncertainty quantification

Other Geotechnical Applications:

- Consolidation settlement
- Creep in clays
- Cyclic degradation
- Strain softening

Key Insight: Many geotechnical problems involve history-dependent behavior → LSTM applications!

References

1. Choi, Y., and Kumar, K. (2023). "A Machine Learning Approach to Predicting Pore Pressure Response in Liquefiable Sands under Cyclic Loading." *Geo-Congress 2023*.
2. Hochreiter, S., and Schmidhuber, J. (1997). "Long Short-Term Memory." *Neural Computation*, 9(8), 1735-1780.
3. Kwan, W. S., Sideras, S. S., Kramer, S. L., and El Mohtar, C. (2017). "Experimental Database of Cyclic Simple Shear Tests under Transient Loadings." *Earthquake Spectra*, 33(3), 1219-1239.
4. Boulanger, R. W., and Ziotopoulou, K. (2017). "PM4Sand (Version 3.1): A sand plasticity model for earthquake engineering applications."

Code: `docs/02-lstm/lstm-liquefaction.ipynb`